

Operators

Operators are special symbols that perform operations on variables and values.

Types of Operators

1. Arithmetic/ Mathematical Operator
2. Relational/ Comparison Operator
3. Logical Operator
4. Assignment Operator
5. Ternary/ Conditional Operator
6. Membership Operator
7. Identity Operator

Arithmetic Operator

Operator	Meaning	Example	Result
+	Addition	10 + 5	15
-	Subtraction	10 - 5	5
*	Multiplication	10 * 5	50
/	Division	10 / 5	2.0
%	Modulus (Remainder)	10 % 3	1
**	Exponent (Power)	2 ** 3	8
//	Floor Division	10 // 3	3

Relational Operator

Operator	Meaning	Example	Result
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	7 > 3	True
<	Less than	2 < 8	True
>=	Greater than or equal to	5 >= 5	True
<=	Less than or equal to	4 <= 6	True

Logical Operator

Operator	Meaning	Example	Result
and	Returns True if both are True	$(5 > 3)$ and $(8 > 6)$	True
or	Returns True if at least one is True	$(5 > 10)$ or $(8 > 6)$	True
not	Reverses the result	not $(5 > 3)$	False

Assignment Operator

Operator	Meaning	Example	Same As
=	Assign	x = 5	—
+=	Add and assign	x += 3	x = x + 3
-=	Subtract and assign	x -= 3	x = x - 3
*=	Multiply and assign	x *= 3	x = x * 3
/=	Divide and assign	x /= 3	x = x / 3
%=	Modulus and assign	x %= 3	x = x % 3

Ternary Operator

```
value_if_true if condition else value_if_false
```

```
1  a=10
2  b=20
3
4  max = a if a>b else b
5  print(max)
6
```

Membership Operator

Operator	Meaning	Example	Result
in	Returns True if present	'a' in 'apple'	True
not in	Returns True if not present	'x' not in 'apple'	True

Identity Operator

Operator	Meaning	Example	Result
is	True if both refer to same object	x is y	True or False
is not	True if not same object	x is not y	True or False

Great Pro8